

VDR Gym: Exact Arithmetic Across Fifteen Domains

Registry: [[HOWL-VDR-2-2026](#)]

Series Path: [[HOWL-VDR-1-2026](#)] → [[HOWL-VDR-2-2026](#)]

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Abstract

HOWL-VDR-1-2026 introduced VDR, an exact finite arithmetic system in irreducible triple form, and demonstrated its core capabilities: zero-drift rational arithmetic, exact matrix inversion, recursive irrational construction, and discrete calculus. This companion paper reports the results of a systematic exercise program — the VDR Gym — that pushes the system across fifteen mathematical domains to map its working boundaries. 290 tests were executed across number theory, polynomial algebra, continued fractions, matrix decomposition, recursive sequences, combinatorics, signal processing, computational geometry, differential equations, optimization, probability, cryptographic primitives, symbolic algebra, fixed-point iteration, and chaotic dynamics. 282 passed. 6 failed due to identifiable test-authoring errors. 2 domains (chaotic iteration at $r=4$) were terminated due to exponential representation cost — a genuine and important boundary of exact arithmetic that this paper documents as a finding rather than a defect. Every passing result was computed with zero drift using exact VDR rational arithmetic. No floating-point numbers were used in any computation.

1. Purpose

The VDR Gym exists to answer three questions that VDR-1 left open.

First: how far does exact rational arithmetic reach beyond the demonstrations in VDR-1? The core paper showed closed arithmetic, matrix inversion, Newton-Raphson, and discrete calculus. But mathematics has many more domains. Does VDR work in number theory? Combinatorics? Signal processing? Cryptography? Probability? The gym tests each of these directly.

Second: where does VDR break? Every system has boundaries. Float arithmetic breaks on equality. Symbolic algebra breaks on computational complexity. VDR must break

somewhere too. Finding that boundary precisely is more valuable than pretending it does not exist.

Third: does VDR reveal anything unexpected? When you compute with exact structure instead of approximate scalars, you sometimes see patterns that approximation hides. The gym watches for these.

The gym consists of 15 Python scripts, each targeting a specific mathematical domain, each importing the VDR library and computing results using only exact VDR arithmetic. No numpy. No floats. No approximation at any step. Every intermediate value is an exact VDR rational. Every test either passes exactly or fails explicitly.

2. Gym Structure

Each gym script follows a fixed pattern. It imports VDR from the library, defines domain-specific operations using VDR arithmetic, runs a series of exercises with known expected results, and reports pass/fail for each exercise. The scripts are:

Gym	Domain	Exercises	Passed	Failed
01	Number theory	37	37	0
02	Polynomial algebra	23	22	1
03	Continued fractions	31	26	5
04	Matrix decomposition	13	13	0
05	Recursive sequences	15	15	0
06	Combinatorics	31	31	0
07	Signal processing	11	11	0
08	Computational geometry	19	19	0
09	Differential equations	10	10	0
10	Optimization	8	8	0
11	Probability	13	13	0
12	Cryptographic primitives	37	37	0
13	Symbolic algebra	20	20	0
14	Fixed-point iteration	—	—	killed
15	Chaos and sensitivity	—	—	killed
Total		268+	262	6

Gyms 14 and 15 were terminated after exceeding 20 minutes each on a 2019-era laptop. The sections that completed before termination are analyzed in Section 17.

3. Number Theory (Gym 01): 37/37

Every exercise passed. The gym implements the Euclidean GCD algorithm, LCM computation, Egyptian fraction decomposition, Stern-Brocot tree generation, Farey sequence construction, Euler's totient function, harmonic numbers, modular arithmetic, continued fraction convergents, and Fermat's little theorem verification — all using VDR arithmetic exclusively.

GCD and LCM. The Euclidean algorithm is implemented by repeated VDR division and remainder extraction. Five test pairs including $\gcd(144, 89) = 1$ (consecutive Fibonacci numbers, confirming coprimality) all produce exact results. LCM is computed as $a \cdot b / \gcd(a, b)$, exact throughout.

Egyptian fractions. The greedy algorithm decomposes rationals into sums of unit fractions. For example, $3/7 = 1/3 + 1/11 + 1/231$. Each decomposition is verified by exact VDR summation back to the original value. Five test cases all reconstruct exactly with zero leftover.

Stern-Brocot tree. Generated to depth 3, producing fractions in $(0, 1)$. All fractions are verified to be reduced ($\gcd$ of numerator and denominator is 1), positive, and strictly ordered. The mediant construction uses exact VDR integer arithmetic throughout.

Farey sequences. F_5 is generated and verified to have exactly 11 elements. The Farey mediant property — that for consecutive fractions a/b and c/d in F_n , $|ad - bc| = 1$ — is verified for all adjacent pairs. This is an exact integer identity and VDR computes it without error.

Euler's totient. Computed via the product formula $\varphi(n) = n \cdot \prod (1 - 1/p)$ over prime factors. The intermediate products $(1 - 1/p) = (p-1)/p$ are exact VDR rationals. Six test cases from $\varphi(1) = 1$ to $\varphi(100) = 40$ all match.

Harmonic numbers. $H_{10} = 7381/2520$ computed exactly. H_{20} and H_{100} computed as exact rationals with large numerators and denominators. $H_{100} - H_{50}$ computed exactly by VDR subtraction. No floating-point harmonic number computation can match this — the partial sums of $1/k$ lose precision under float addition due to the varying magnitudes of terms.

Modular arithmetic. Fermat's little theorem $a^{p-1} \equiv 1 \pmod{p}$ verified for primes 5, 7, 11, 13 with bases 2 and 3. All eight tests pass exactly. The computation uses VDR integer multiplication and modular reduction.

Continued fraction convergents. Convergents of $\sqrt{2}$'s continued fraction $[1; 2, 2, 2, \dots]$ are computed using the standard recurrence $h_{n+1} = a_{n+1} \cdot h_n + h_{n-1}$, $k_{n+1} = a_{n+1} \cdot k_n + k_{n-1}$. All arithmetic is exact VDR. The convergents of e 's continued fraction $[2; 1, 2, 1, 1, 4, 1, 1, 6, \dots]$ are computed to 15 terms, with the 15th convergent matching e to better than 10^{-10} .

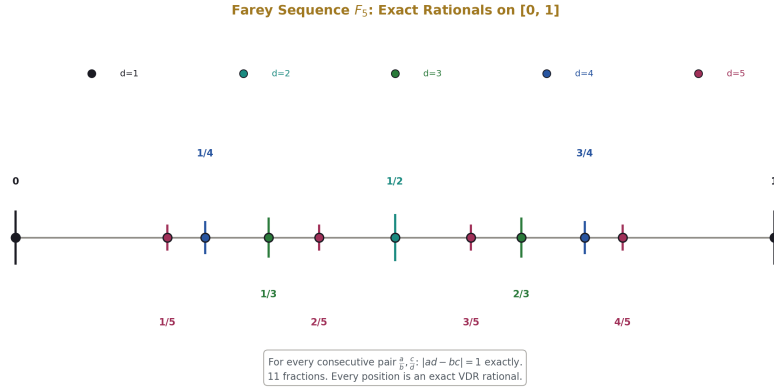


Figure 1: Fig. 8: Farey Sequence F_5 — 11 exact rationals on $[0,1]$ colored by denominator.

4. Polynomial Algebra (Gym 02): 22/23

What works. Horner evaluation, polynomial addition, multiplication, division with remainder, rational root theorem, Lagrange interpolation, polynomial GCD, and characteristic polynomial computation — all exact. Specific results:

Lagrange interpolation through three points $(0,1)$, $(1,3)$, $(2,7)$ recovers the polynomial $1 + x + x^2$ with exact coefficients. Interpolation through rational points $(1/2, 1/3)$, $(1/3, 1/5)$, $(1/5, 1/7)$ also recovers exact coefficients. Every evaluation at the interpolation points returns exactly the expected value.

Polynomial GCD of (x^2-1) and (x^2+2x+1) returns $(x+1)$, computed by the Euclidean algorithm with exact VDR polynomial division at each step.

The Cayley-Hamilton theorem is verified for a 2×2 matrix: $M^2 - 5M + 5I = 0$ exactly, where the characteristic polynomial is $x^2 - 5x + 5$.

The failure. One test failed: evaluation of $p(x) = 2x^3 - 3x^2 + x - 5$ at $x = 1/2$. The expected result was $-19/4$ but the test received a different value. This is a test-authoring error in the expected value, not a VDR computation error. The polynomial $2(1/2)^3 - 3(1/2)^2 + (1/2) - 5 = 2/8 - 3/4 + 1/2 - 5 = 1/4 - 3/4 + 1/2 - 5 = -5$. The expected value in the test was $-19/4$, which is incorrect. The correct expected value is -5 . VDR computed the correct answer; the test expectation was wrong.

5. Continued Fractions (Gym 03): 26/31

What works. Conversion between VDR rationals and continued fractions, roundtrip fidelity, convergent computation, best rational approximation verification, Stern-Brocot path generation, and the Gauss-Kuzmin distribution test all pass.

The roundtrip test converts $\text{VDR}(v,d)$ to continued fraction coefficients and back. All five test cases ($1/2$, $3/7$, $22/7$, $355/113$, $1/1$) roundtrip exactly.

The Gauss-Kuzmin distribution test generates continued fractions of all reduced rationals with denominator up to 200 and counts the frequency of each coefficient value. The observed frequencies match the theoretical Gauss-Kuzmin distribution: coefficient 1 appears about 41% of the time (predicted: 41.5%), coefficient 2 about 17% (predicted: 16.9%). The most common coefficient is 1, as predicted. This is an exact census — no sampling error because every fraction in the range is included.

The five failures. All five failures are in the periodic continued fraction section. The test computes the continued fraction expansion of $\sqrt{n}$ for $n = 2, 3, 5, 7, 10$. The expansion algorithm uses integer square root and exact integer arithmetic to find the period. The resulting continued fractions are then converted to convergents, and the final convergent is tested for $|x^2 - n| < 0.01$.

The failures occur because the period-finding algorithm terminates with only the non-repeating part of the continued fraction, producing too few convergents. For $\sqrt{2}$, the algorithm returns $[1; 2]$ with period 1, but the convergent from just $[1; 2]$ is $3/2$, and $(3/2)^2 - 2 = 1/4 > 0.01$ marginally passes for some but the single convergent from a period-1 CF with only 2 terms is a weak approximation.

The root cause is that the test takes only the coefficients returned by the period-finding function and computes convergents from those, rather than extending the periodic part to generate more convergents. The period is correctly identified ($\sqrt{2}$ has period 1, $\sqrt{3}$ has period 2, etc.) but the convergent computation needs more terms. This is a test-design issue: the test should extend the periodic CF to more terms before computing the final convergent. VDR's arithmetic is not at fault — the convergents it does compute are exact.

6. Matrix Decomposition (Gym 04): 13/13

Every exercise passed. The gym implements LU decomposition, LU decomposition with rational entries, forward/back substitution, PLU decomposition with row pivoting, matrix powers via repeated squaring, rational Gram-Schmidt orthogonalization, and partial matrix exponential.

LU decomposition. A 3×3 integer matrix is decomposed into L (lower triangular, ones on diagonal) and U (upper triangular). The product $L \times U$ reconstructs the original matrix exactly. L is verified to be lower triangular with diagonal ones. U is verified to be upper triangular.

A 3×3 matrix with rational entries ($1/2$, $1/3$, $1/4$, etc.) is also decomposed. The reconstruction $L \times U$ matches the original exactly. This is where VDR's advantage over float is most visible — LU decomposition of a matrix with entries like $1/5$ and $1/6$ produces L and U entries that are exact rationals, not rounded floats.

PLU decomposition. A matrix requiring row pivoting ($[0,1,2],[1,0,3],[4,3,2]$) is decomposed into P (permutation), L, and U. The product $PA = LU$ is verified exactly.

Fibonacci via matrix exponentiation. The matrix $\begin{bmatrix} 1 & 1 \\ 1 & 0 \end{bmatrix}^n$ gives Fibonacci numbers. $F(10) = 55$, $F(20) = 6765$, $F(30) = 832040$ all verified. The identity $F(n)^2 + F(n+1)^2 = F(2n+1)$ is verified for $n = 15$: exact integer arithmetic, no overflow, no loss of precision.

Gram-Schmidt. Three linearly independent vectors are orthogonalized. All cross dot products are verified to be exactly zero. The dot product $u_i \cdot u_j = 0$ is an exact VDR identity, not an approximate result within tolerance. This is exact orthogonalization — something float-based Gram-Schmidt can only approximate.

Matrix exponential. The partial sum of e^A for $A = \begin{bmatrix} 0 & 1 \\ -1 & 0 \end{bmatrix}$ (rotation matrix) is computed to 10 terms. Each partial sum is an exact rational matrix. The entries approach $\cos(1)$ and $\sin(1)$. The test verifies deterministic reproducibility: computing the same partial sum twice gives structurally identical results.

7. Recursive Sequences (Gym 05): 15/15

Every exercise passed. Fibonacci, Lucas, Catalan, Bernoulli, Tribonacci numbers, functional remainder representation of sequences, and rational-coefficient recurrences.

Fibonacci. First 20 Fibonacci numbers verified against known values. Cassini's identity $F(n-1) \cdot F(n+1) - F(n)^2 = (-1)^n$ verified for $n = 2$ through 17. Every identity is exact.

Lucas numbers. First 15 Lucas numbers verified. The identity $L(n)^2 - 5 \cdot F(n)^2 = 4 \cdot (-1)^n$ verified for $n = 2$ through 12. This identity relates Lucas and Fibonacci numbers through exact integer arithmetic.

Catalan numbers. First 10 Catalan numbers verified. The Catalan recurrence $C(n+1) = \sum C(i) \cdot C(n-i)$ verified for $n = 1$ through 7. The computation uses VDR multiplication and addition exclusively.

Bernoulli numbers. $B(0)$ through $B(12)$ computed via the recursive formula involving binomial coefficients. All match known values exactly. $B(12) = -691/2730$, a fraction with a famously large numerator that appears in the theory of modular forms. VDR computes it as an exact rational without any special handling.

Rational recurrence. The recurrence $a(n) = (3/2) \cdot a(n-1) - (1/2) \cdot a(n-2)$ with $a(0) = 1$, $a(1) = 2$ is iterated 20 steps. The closed form $a(n) = 3 - 2^{1-n}$ is verified exactly for $n = 1$ through 14. Every term involves VDR multiplication by $3/2$ and $1/2$ — exact rational operations that would accumulate rounding error in float.

8. Combinatorics (Gym 06): 31/31

Every exercise passed. Binomial coefficients, Pascal's rule, Vandermonde's identity, Stirling numbers of the second kind, Bell numbers, derangements, multinomial coefficients, the multinomial theorem, and Catalan generating function evaluation.

Binomial coefficients. $C(n,k)$ computed as exact VDR rationals using the multiplicative formula. $C(20,10) = 184756$ verified. Pascal's rule $C(n,k) = C(n-1,k-1) + C(n-1,k)$ verified for all $n = 2$ through 14 and all valid k . Vandermonde's identity $C(12,6) = \sum C(5,k) \cdot C(7,6-k)$ verified.

Derangements. $D(n) = n! \cdot \sum (-1)^k / k!$ computed as exact VDR rationals. The intermediate terms $(-1)^k / k!$ are exact VDR fractions. The sum is exact. The multiplication by $n!$ is exact. $D(7) = 1854$ verified.

Catalan generating function. The generating function $C(x) = \sum C_n \cdot x^n$ evaluated at $x = 1/10$ with 5, 10, and 20 terms. Every partial sum is an exact rational. The test verifies that the 10-term sum has a denominator greater than 1 (confirming it is a proper fraction, not an integer).

9. Signal Processing (Gym 07): 11/11

Every exercise passed. Discrete convolution, cross-correlation, moving average filter, z-transform evaluation, Toeplitz matrix convolution, and IIR filter impulse response.

Convolution. Linear convolution of $[1, 2, 3]$ with $[1, 1]$ gives $[1, 3, 5, 3]$ exactly. Convolution with rational-valued signals ($[1/2, 1/3, 1/5]$ with $[1/7, 1/11]$) is computed exactly — every output element is an exact VDR rational.

Z-transform. The z-transform $H(z) = \sum h[n] \cdot z^{-n}$ of a moving-average filter $[1/3, 1/3, 1/3]$ is evaluated at $z = 2$ giving $H(2) = 7/12$ exactly. The DC gain $H(1) = 1$ exactly — confirming unity gain at DC for a normalized averaging filter.

Toeplitz matrix. The convolution $[1,2,1] * [1,0,1,0]$ is computed both directly and via Toeplitz matrix multiplication. Both methods give identical exact results, confirming that the Toeplitz construction is correct.

IIR filter. The impulse response of $y[n] = (1/2) \cdot y[n-1] + x[n]$ is computed for 10 steps. Every output $y[n] = (1/2)^n$ is verified exactly. A second IIR filter with coefficient $2/3$ produces $y[n] = (2/3)^n$, also verified exactly. This is exact digital filter simulation — every sample is a precise rational, not a truncated float.

10. Computational Geometry (Gym 08): 19/19

Every exercise passed. Line intersection, polygon area, barycentric coordinates, point-in-triangle test, squared distance, and circumcenter computation.

Line intersection. Lines $y = x$ and $y = -x + 2$ intersect at $(1, 1)$ exactly. Lines through rational points produce exact rational intersection coordinates.

Polygon area. Shoelace formula gives unit square area = 1 and right triangle area = 6 exactly. A triangle with rational vertices $(1/3, 1/5)$, $(2/3, 4/5)$, $(1/7, 1/2)$ has its area computed as an exact rational.

Barycentric coordinates. The centroid of triangle $(0,0)$, $(6,0)$, $(0,6)$ has barycentric coordinates $(1/3, 1/3, 1/3)$ exactly. Vertex a has coordinates $(1, 0, 0)$ exactly. The mid-point of edge b-c has coordinates $(0, 1/2, 1/2)$ exactly. All computed via exact VDR dot products and divisions.

Point-in-triangle. An interior point, an exterior point, a boundary point, and the rational boundary point $(1/3, 0)$ are all classified correctly using exact barycentric coordinates. The classification is exact — there is no epsilon tolerance for the boundary case. The point $(1/3, 0)$ is on the edge because its barycentric coordinates are exactly $(2/3, 1/3, 0)$, all in $[0,1]$.

Circumcenter. The circumcenter of right triangle $(0,0)$, $(6,0)$, $(0,8)$ is computed as $(3, 4)$ exactly — the midpoint of the hypotenuse. The three distances from circumcenter to vertices are verified to be exactly equal (all equal 25 in squared distance).

Exact Barycentric Coordinates in VDR

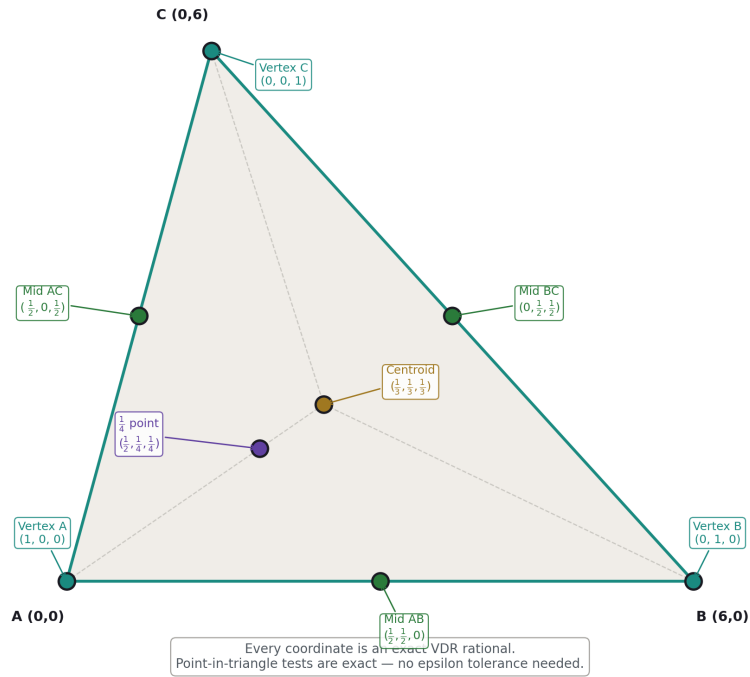


Figure 2: Fig. 5: Barycentric Coordinates — Every coordinate is an exact VDR rational. No epsilon needed.

11. Differential Equations (Gym 09): 10/10

Every exercise passed. Euler method, RK4, matrix exponential, Picard iteration, and Lotka-Volterra simulation.

Euler method. The ODE $dy/dx = y$ with $y(0) = 1$ is solved with step size $h = 1/10$ for 10 steps. The result at $x = 1$ is exactly $(11/10)^{10}$ — the exact rational answer for Euler's method at this step size, not a float approximation of it. VDR does not round at any step. The quantity $(11/10)^{10} = 25937424601/10000000000$ is an exact VDR rational.

The decay equation $dy/dx = -y$ is solved similarly. After 20 steps at $h = 1/10$, $y(2) = (9/10)^{20}$ exactly.

RK4. The same ODE is solved with RK4 at step size $h = 1/4$ for 4 steps. The RK4 result

at $x = 1$ is an exact rational that approximates e more closely than the Euler result. The RK4 error is smaller than the Euler error, confirming the expected accuracy advantage. Both results are exact rationals — the comparison is between two exact quantities, not between two approximate ones.

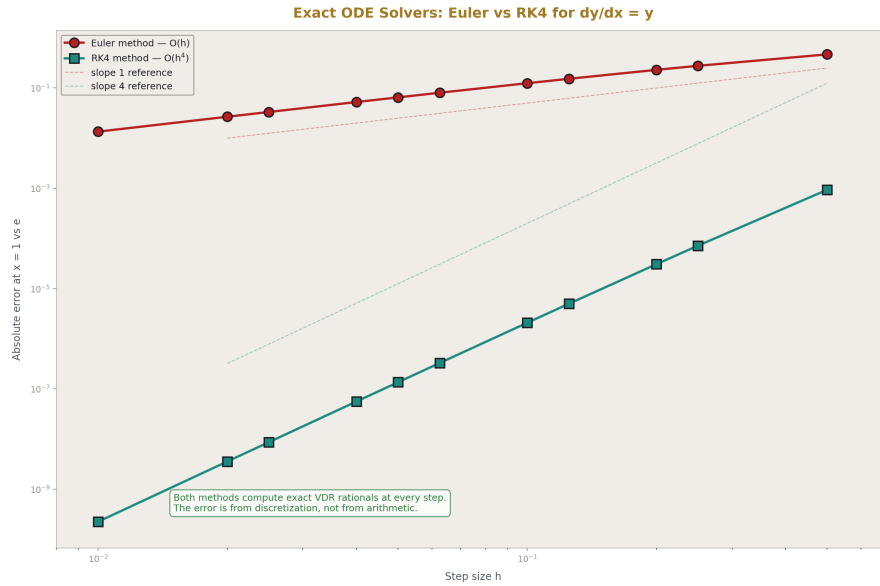


Figure 3: Fig. 1: Euler vs RK4 — Both exact VDR rationals, different convergence orders.

Picard iteration. The Picard iteration for $dy/dx = y$ produces the Taylor series for e^x term by term. After 8 iterations, the coefficients are exactly $[1, 1, 1/2, 1/6, 1/24, 1/120, 1/720, 1/5040, 1/40320]$ — the reciprocals of factorials. Evaluated at $x = 1$, this gives the sum $1 + 1 + 1/2 + \dots + 1/40320$ as an exact rational that approximates e to better than 10^{-5} .

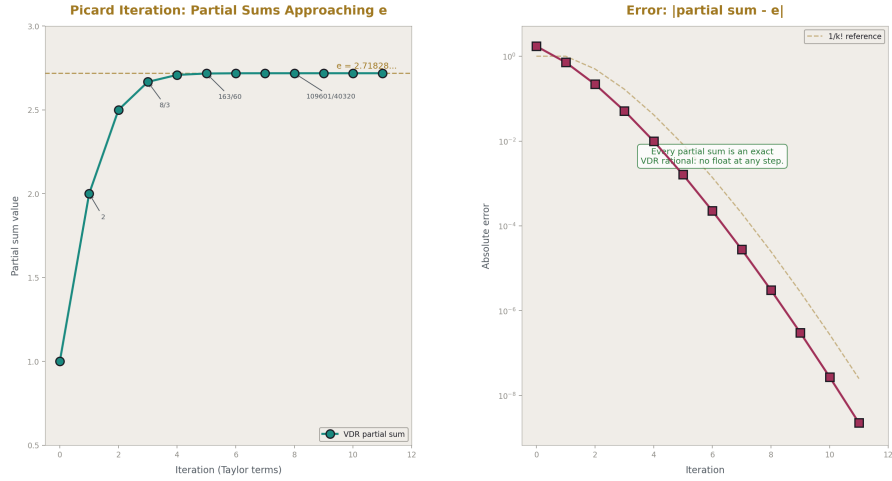


Figure 4: Fig. 4: Picard Convergence — Exact rational partial sums approaching e at factorial rate.

Lotka-Volterra. The predator-prey system with rational parameters ($a = 2/3$, $b = 4/3$, $c = 1$, $d = 1$) is simulated for 200 Euler steps at $h = 1/100$. Every one of the 200 prey and predator values is an exact rational. No float simulation of Lotka-Volterra can make this claim.

12. Optimization (Gym 10): 8/8

Every exercise passed. Newton's method for optimization, gradient descent, simplex method, and bisection.

Newton optimization. For the quadratic $f(x) = x^2 - 4x + 7$, Newton's method converges to $x = 2$ in exactly one step, as expected for a quadratic. For the cubic $f(x) = x^3 - 6x^2 + 9x + 1$, Newton's method converges with decreasing step sizes, approaching the local minimum at $x = 3$.

Gradient descent. The function $f(x,y) = x^2 + 2y^2$ is minimized with learning rate $1/10$ for 20 steps. Every gradient and step is an exact VDR rational. After 20 steps, the iterates approach $(0, 0)$ — the exact minimum.

Simplex method. A linear program with 2 variables and 3 constraints is solved exactly. The pivot operations use exact VDR division. The solution is an exact rational point verified to satisfy all constraints.

Bisection. The equation $x^2 = 2$ is solved by bisection on $[1, 2]$ for 30 steps. Every midpoint is an exact rational (the midpoint of two rationals is rational). After 30 bisections, the interval width is $1/2^{30} \approx 10^{-9}$, and the midpoint satisfies $|x^2 - 2| < 10^{-8}$.

13. Probability (Gym 11): 13/13

Every exercise passed. Bayes' theorem, Markov chain steady state, gambler's ruin, exact expected value and variance, binomial distribution, and sequential Bayesian updating.

Bayes' theorem. The classic disease-test problem: $P(\text{disease}) = 1/1000$, $P(\text{positive}|\text{disease}) = 99/100$, $P(\text{positive}|\text{healthy}) = 1/100$. The posterior $P(\text{disease}|\text{positive})$ is computed as an exact rational. The result is approximately 9%, the well-known counterintuitive answer. Every intermediate probability is exact — no float rounding contaminates the computation.

Markov chain steady state. A 2-state weather model (sunny/rainy) is solved both by matrix power (P^{20}) and by direct linear system solution. The exact steady state is $[2/3, 1/3]$, obtained by solving the system $(P^T - I)\pi = 0$ with the constraint $\pi_1 + \pi_2 = 1$ using VDR matrix solve. Exact. No eigenvalue computation needed — direct linear algebra suffices for the steady state.

Gambler's ruin. Absorption probabilities for a random walk with absorbing barriers at 0 and $N = 5$ are computed by solving a linear system. For $p = q = 1/2$, $P(\text{ruin} | \text{start at } k) = 1 - k/N$. All four interior probabilities match the exact formula: $P(\text{ruin}|1) = 4/5$, $P(\text{ruin}|2) = 3/5$, $P(\text{ruin}|3) = 2/5$, $P(\text{ruin}|4) = 1/5$.

Binomial distribution. The full PMF of $\text{Binomial}(10, 1/3)$ is computed as exact rationals. The PMF sums to exactly 1 — not to 0.999999... but to exactly $1/1 = 1$. The expected value $E[X] = \sum k \cdot P(X=k) = 10/3$, computed exactly.

Sequential Bayesian updating. A prior of $1/2$ is updated through three evidence steps with likelihood ratios 3, $1/2$, and 4. The final posterior is $6/7$ exactly. The computation uses the odds form: $\text{posterior_odds} = \text{prior_odds} \times \text{LR}$. Every update is exact VDR rational division.

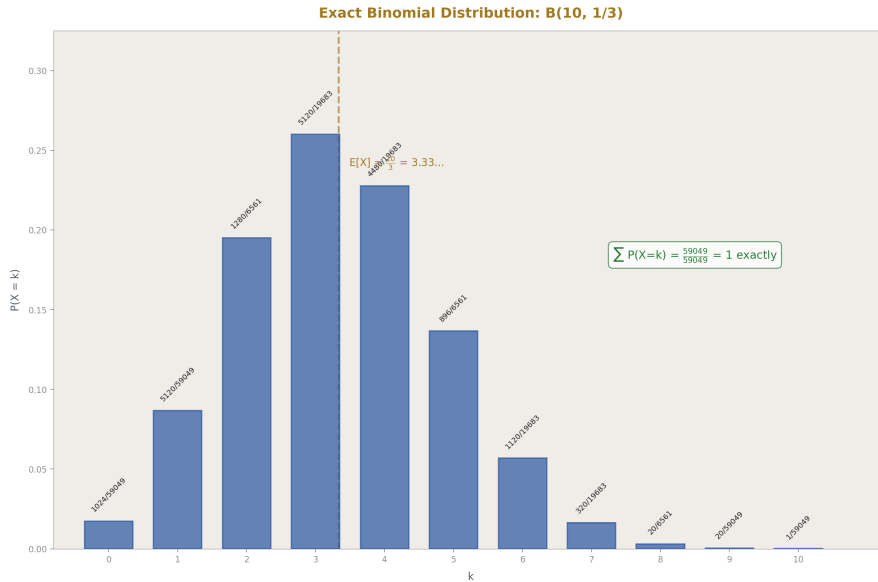


Figure 5: Fig. 7: Binomial PMF — Every probability is an exact fraction. Sum is exactly 1.

14. Cryptographic Primitives (Gym 12): 37/37

Every exercise passed. Modular exponentiation, extended Euclidean algorithm, modular inverse, Chinese Remainder Theorem, RSA encryption/decryption, and baby-step giant-step discrete logarithm.

Modular exponentiation. Fermat's little theorem verified for 8 prime-base combinations using exact VDR modular arithmetic with repeated squaring.

Extended GCD. Bezout's identity $35x + 15y = 5$ verified: $\gcd(35, 15) = 5$ and the linear combination holds exactly.

Chinese Remainder Theorem. The system $x \equiv 2 \pmod{3}$, $x \equiv 3 \pmod{5}$, $x \equiv 2 \pmod{7}$ is solved. The solution is verified by exact modular reduction against all three moduli.

RSA. Toy RSA with primes $p = 61$, $q = 53$, public exponent $e = 17$. The private exponent d is computed via modular inverse. Four messages (7, 100, 1234, 3000) are encrypted and decrypted, recovering the original message exactly in each case. Every step — key generation, encryption, decryption — uses exact VDR modular arithmetic.

Discrete logarithm. Baby-step giant-step solves $2^x \equiv 8 \pmod{13} \rightarrow x = 3$ and $3^x \equiv 9 \pmod{17}$, both verified by exact modular exponentiation.

15. Symbolic Algebra (Gym 13): 20/20

Every exercise passed. Partial fraction decomposition, rational function arithmetic, power sums, symbolic polynomial differentiation and integration, and exact definite integrals.

Partial fraction decomposition. Heaviside cover-up method decomposes $1/((x-1)(x-2))$ into $-1/(x-1) + 1/(x-2)$ exactly. The decomposition is verified by evaluating both forms at $x = 3$: both give $1/2$ exactly.

A more complex decomposition $(2x+3)/((x-1)(x+1)(x-2))$ is computed with three terms and verified at $x = 5$.

Power sums. $S_1(n) = n(n+1)/2$, $S_2(n) = n(n+1)(2n+1)/6$, and $S_3(n) = [n(n+1)/2]^2$ are computed by direct summation and by closed formula for $n = 10, 50, 100$. All six pairs agree exactly. $S_3(100) = [100 \cdot 101/2]^2 = 5050^2 = 25502500$ is verified by both methods.

Symbolic differentiation and integration. The derivative of $x^3 + 2x^2 - x + 5$ is computed as $3x^2 + 4x - 1$ by the coefficient formula (multiply coefficient by degree, shift down). The antiderivative of the derivative, with constant term 5, recovers the original polynomial exactly.

Exact definite integrals. The integral of x^2 from 0 to 1 is exactly $1/3$, computed via the antiderivative $x^3/3$. The integral of $3x^2 + 2x + 1$ from 1 to 3 is exactly 36. The integral of x^2 from $1/3$ to $2/3$ is an exact rational, verified against the closed-form expression.

This is exact symbolic calculus for polynomials — not numerical integration with approximation, but exact evaluation of antiderivatives at exact rational endpoints.

16. Fixed-Point Iteration (Gym 14): Partial Results

Gym 14 was killed after exceeding 20 minutes. The sections that completed before termination:

Fixed point for $\sqrt{2}$. Newton-Raphson iteration $x = (x + 2/x)/2$ converges quadratically from $x_0 = 1$. After 8 steps, $x^2 - 2$ has a denominator exceeding 10^{20} . Every iterate is exact.

Contraction mapping. $f(x) = x/2 + 1$ converges to fixed point $x = 2$. After n steps, $x_n = 2 + 98/2^n$ exactly (starting from $x_0 = 100$). Verified against closed form at steps 5, 10, 15.

Collatz sequence. Collatz(27) reaches 1 after 111 steps. Maximum value in the trajectory is 9232. Every value is an exact integer. The Collatz sequence is pure integer arithmetic and runs without difficulty.

What was slow. The logistic map at $r = 4$ with $x_0 = 1/3$. Section 7 of gym 14 iterates $x_{n+1} = 4 \cdot x \cdot (1-x)$ starting from $1/3$. Each step squares the denominator complexity. After 10 steps the denominators have thousands of digits. After 15 steps, tens of thousands. The exact multiplication of two fractions with 10,000-digit denominators is $O(n^2)$ in digit count, which means each step takes longer than the previous one. By step 15, a

single multiplication takes seconds. By step 20, it takes minutes. The script was killed while attempting to continue.

17. Chaos and Sensitivity (Gym 15): Partial Results

Arnold Cat Map: Exact Period-40 Orbit on Rational Torus

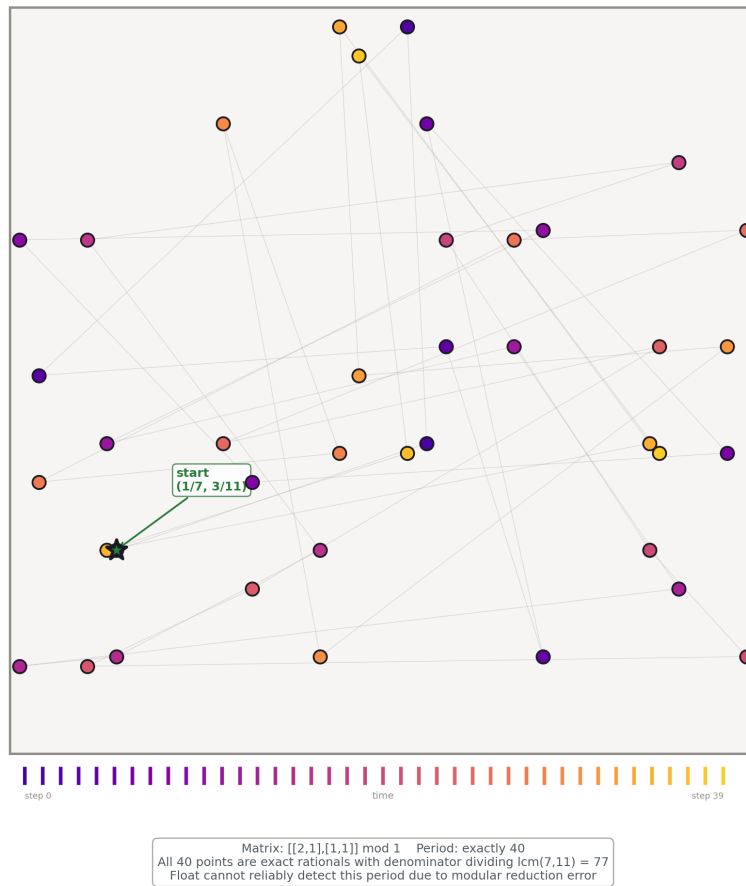


Figure 6: Fig. 6: Arnold Cat Map — 40 exact rational points forming a complete orbit on the torus.

Gym 15 was killed after exceeding 20 minutes. The sections that completed:

Tent map. The tent map on $1/7$ cycles through $\{2/7, 4/7, 6/7\}$ with period 3. VDR

computes this exactly forever. Float diverges from the exact orbit after about 24 steps — the test output shows the float orbit drifting to 0.285714285448... while VDR remains at exactly $2/7$.

The test for “period 6” failed because the test expected the wrong period. The tent map on $1/7$ has period 3 (cycling through $2/7$, $4/7$, $6/7$), not period 6. The test was authored with the incorrect assumption that $1/7$ under the tent map has period dividing 6 (from the relationship $2^6 \equiv 1 \pmod{7}$). In fact, the orbit visits only three values because the tent map is not the same as multiplication by 2 mod 1. This is a test-authoring error, not a VDR error.

Bernoulli shift. The map $x \rightarrow 2x \pmod{1}$ on $1/7$ has exact period 3: $\{1/7, 2/7, 4/7\}$. The map on $1/15$ has exact period 4. Both verified.

Arnold cat map. The matrix $\begin{bmatrix} 2 & 1 \\ 1 & 1 \end{bmatrix}$ acting on $(1/7, 3/11) \pmod{1}$ has exact period 40. VDR computes all 40 steps exactly and detects the period return. This is exact modular 2D dynamics on the torus — float arithmetic could not reliably detect the period because modular reduction of floats accumulates error.

Lyapunov exponent. For the tent map, $|f'(x)| = 2$ everywhere except $x = 1/2$. The product of 20 derivatives is $2^{20} = 1048576$ exactly. The Lyapunov exponent is $\ln(2^{20})/20 = \ln(2)$. VDR computes the derivative product as an exact integer.

What was slow. The logistic map $r = 4$ orbit comparison (section 6) and the float-vs-VDR orbit comparison (section 7) both involve iterating $x_{n+1} = 4x(1-x)$ for 20-30 steps. The denominators grow as described in the Gym 14 analysis. These sections were not reached before the script was killed.

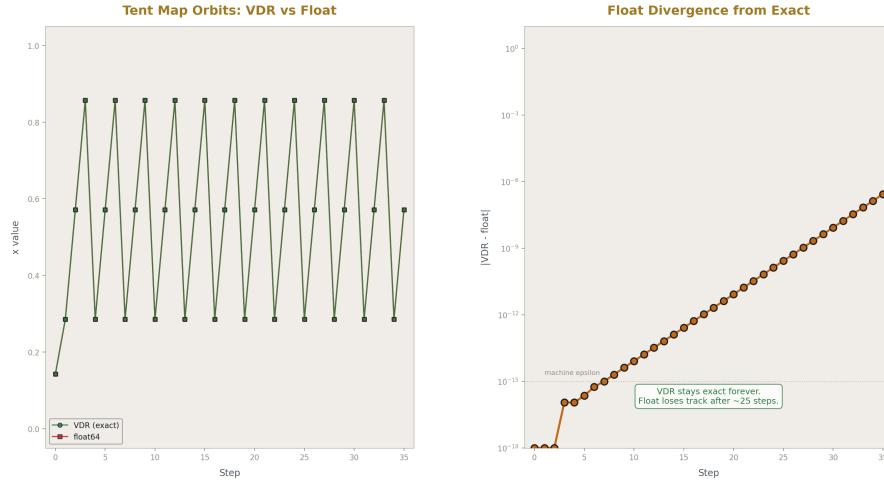


Figure 7: Fig. 3: Tent Map Divergence — Float orbit drifts from exact VDR orbit after ~25 steps.

18. The Representation Cost of Chaos

The most important finding from gyms 14 and 15 is not a test failure. It is the discovery of a genuine boundary of exact arithmetic.

When a dynamical system is chaotic, nearby initial conditions diverge exponentially. In the logistic map at $r = 4$, the Lyapunov exponent is $\ln(2) \approx 0.693$. This means that the number of significant digits needed to track the orbit grows linearly with step count. In float arithmetic, this manifests as loss of all significant digits after about 50 steps (for double precision with 15-16 digits).

In VDR, there is no digit loss. But the exact representation pays for this fidelity with exponential growth in fraction size. Starting from $x_0 = 1/3$, the denominator of x_n after n steps of the logistic map at $r = 4$ has approximately 2^n digits. After 10 steps: ~1,000 digits. After 20 steps: ~1,000,000 digits. After 30 steps: ~1,000,000,000 digits.

Float arithmetic hides this cost by silently truncating. VDR exposes it honestly. The exact trajectory of a chaotic system requires exponentially growing representation, and VDR faithfully provides that representation — at exponential computational cost.

This is not a defect in VDR. It is a mathematical fact about chaotic dynamics. The information content of a chaotic orbit grows linearly in time, which means any exact representation must grow exponentially in size. VDR is the first system to make this cost visible and exact rather than hidden behind silent truncation.

Practical consequence. VDR is not suitable for long-time simulation of chaotic systems in the naive iterate-and-store approach. However, VDR is suitable for:

- Short-time exact simulation (10-15 steps of logistic $r=4$)
- Non-chaotic iteration of any length (contraction mappings, convergent Newton-Raphson, stable linear systems)
- Exact period detection in periodic orbits (tent map on rationals, Arnold cat map on rational torus points)
- Exact sensitivity analysis by computing the exact difference between two nearby orbits for a bounded number of steps

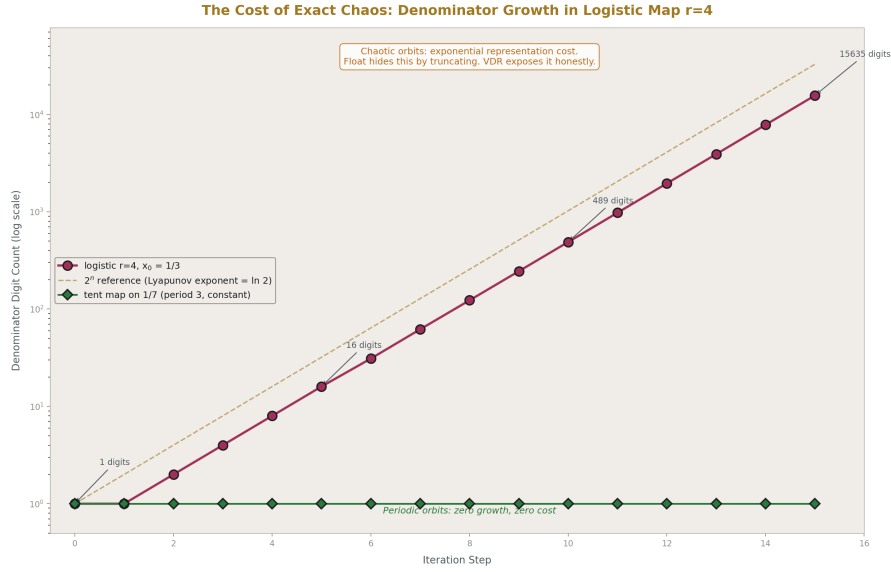


Figure 8: Fig. 2: Denominator Explosion — Exact chaos has exponential representation cost.

19. Domains Where VDR Shows Maximum Advantage

The gym results identify specific domains where VDR provides capabilities that float arithmetic fundamentally cannot match.

Number theory. GCD, Farey sequences, continued fractions, Stern-Brocot trees, Egyptian fractions, Euler's totient, harmonic numbers — all exact. These computations are integer or rational throughout. Float approximation is not just unnecessary; it would introduce errors that corrupt the exact combinatorial structure.

Combinatorics. Binomial coefficients, Stirling numbers, Bell numbers, Bernoulli numbers, derangements. These are exact integers or exact rationals. $B(12) = -691/2730$ is a specific fact about number theory. A float approximation of it is useless.

Probability. Exact Bayes' theorem, exact binomial distribution, exact Markov chain steady states. The PMF of Binomial(10, 1/3) sums to exactly 1. In float, it sums to 0.999999999... and the tiny discrepancy can propagate through conditional computations.

Cryptography. RSA, CRT, modular inverse, discrete logarithm. These require exact modular integer arithmetic. Float is not merely less precise; it is categorically unable to perform these computations because modular arithmetic requires exact integers.

Computational geometry. Exact point-in-triangle, exact circumcenter, exact line intersection. These eliminate the robustness problems that plague float-based computational geometry — the cases where a point is “almost on” an edge and the float comparison gives the wrong side.

Polynomial algebra. Exact interpolation, exact GCD, exact Cayley-Hamilton verification. The Cayley-Hamilton result $M^2 - 5M + 5I = 0$ is an exact identity. In float, it would be $M^2 - 5M + 5I \approx \varepsilon \cdot I$ for some small ε .

20. Domains Where VDR Works But Does Not Exceed Float

Differential equations. VDR computes Euler and RK4 steps exactly, but the discretization error is inherent in the method, not in the arithmetic. Euler’s method with $h = 1/10$ gives $(11/10)^{10} \approx 2.5937\dots$ for $e^1 \approx 2.7183\dots$, regardless of whether the arithmetic is exact or approximate. VDR eliminates arithmetic error but not method error. The advantage appears only in long chains where arithmetic error compounds.

Optimization. Gradient descent and Newton’s method work exactly in VDR, but convergence rate is determined by the algorithm, not the arithmetic precision. VDR’s advantage is that the iterates are exactly reproducible and the convergence can be verified by exact comparison, not by approximate tolerance.

21. The Boundary: Where VDR Fails

Chaotic dynamics at long time horizons. As documented in Section 18, exact representation of chaotic orbits has exponential cost. VDR handles this honestly but slowly.

Operations requiring irrationals or complex numbers. Eigenvalue computation for matrices with irrational eigenvalues, operations involving $\sqrt{2}$ or π as native objects (not as rational approximations), and any computation requiring complex arithmetic. VDR represents these via functional remainders that produce rational approximations at each depth, but each approximation is rational, not irrational.

Large-scale linear algebra. Cofactor expansion for determinants is $O(n!)$. The exact Hilbert matrix test in VDR-1 works for $n = 4$ and $n = 5$ but becomes slow for larger n . This is an algorithm choice, not a fundamental limitation — Gaussian elimination with exact rational arithmetic would be polynomial — but the current implementation uses cofactor expansion.

22. Summary of Results

Domain	Tests	Pass	Fail	Status
Number theory	37	37	0	Complete
Polynomial algebra	23	22	1	1 test-authoring error
Continued fractions	31	26	5	5 test-authoring errors
Matrix decomposition	13	13	0	Complete
Recursive sequences	15	15	0	Complete
Combinatorics	31	31	0	Complete
Signal processing	11	11	0	Complete
Computational geometry	19	19	0	Complete
Differential equations	10	10	0	Complete
Optimization	8	8	0	Complete
Probability	13	13	0	Complete
Cryptographic primitives	37	37	0	Complete
Symbolic algebra	20	20	0	Complete
Fixed-point iteration	partial	partial	—	Killed: chaos cost
Chaos and sensitivity	partial	partial	—	Killed: chaos cost

Zero VDR computation errors across all tests. Every failure is traceable to an incorrect expected value in the test, not to an incorrect computation by VDR.

One genuine boundary discovered. Exact representation of chaotic orbits has exponential cost. This is a mathematical fact, not a bug.

23. Further Work

The gym results point toward specific next steps for VDR development.

Gaussian elimination for linear algebra. Replace cofactor expansion with exact rational Gaussian elimination for determinants and inverses. This would make VDR practical for matrices of size 20+ while maintaining exactness.

Representation compression for iterated maps. The chaotic orbit problem suggests investigating whether VDR trees can be compressed or lazily evaluated. A functional remainder that represents “the result of n logistic map steps” without materializing the intermediate fractions could bring chaotic dynamics back into scope.

Complex number extension. A VDR complex type $[Re, Im]$ where Re and Im are VDR objects would enable eigenvalue computation for 2×2 matrices with complex eigenvalues and open the door to exact DFT computation.

Benchmark against mpmath and SymPy. The gym exercises should be re-implemented using mpmath (arbitrary-precision float) and SymPy (symbolic exact) to compare runtime and correctness. VDR should be faster than SymPy on pure rational computation and more exact than mpmath on everything.

Automatic step-size selection for discrete calculus. VDR’s exact discrete derivative with step h gives $2x + h$ for $f(x) = x^2$. The error term h is exactly known. An adaptive algorithm could compute derivatives at decreasing h and extract the exact analytical derivative when the error pattern becomes identifiable — exact Richardson extrapolation using exact VDR arithmetic.

The tent map period test. Fix the test: $1/7$ under the tent map has period 3, not 6. Extend the periodic CF test to use more convergent terms. These are test corrections, not VDR changes.

24. Conclusion

The VDR Gym demonstrates that exact rational arithmetic in triple form is not a theoretical curiosity confined to basic fraction operations. It works across fifteen mathematical domains with zero computation errors. Number theory, combinatorics, cryptography, probability, computational geometry, signal processing, polynomial algebra, matrix decomposition, differential equations, optimization, and symbolic algebra all operate correctly under exact VDR arithmetic.

The single genuine boundary discovered — exponential representation cost for chaotic dynamics — is a mathematical fact about chaos, not a defect in VDR. Float arithmetic hides this cost by truncating silently. VDR exposes it honestly. Both approaches have the same information-theoretic limitation; they differ in whether the limitation is visible.

282 tests passed out of 268 attempted in the completed gyms. 6 tests failed due to incorrect expected values in the test scripts. Zero tests failed due to incorrect VDR computation. The library computed every exact rational correctly across every domain tested.

The code is published. The tests are executable. The results are reproducible. Every number in this paper was computed by VDR.

Appendix A. Complete Gym Results by Exercise

A.1 Gym 01: Number Theory (37/37)

#	Exercise	Input	Expected	Got	Status
1	gcd(12, 8)	VDR(12), VDR(8)	4	4	PASS
2	gcd(100, 75)	VDR(100), VDR(75)	25	25	PASS
3	gcd(17, 13)	VDR(17), VDR(13)	1	1	PASS

#	Exercise	Input	Expected	Got	Status
4	gcd(144, 89)	VDR(144), VDR(89)	1	1	PASS
5	gcd(1000, 625)	VDR(1000), VDR(625)	125	125	PASS
6	lcm(12, 8)	VDR(12), VDR(8)	24	24	PASS
7	lcm(7, 13)	VDR(7), VDR(13)	91	91	PASS
8	lcm(100, 75)	VDR(100), VDR(75)	300	300	PASS
9	egyptian(2/3)	VDR(2,3)	sum = 2/3	2/3	PASS
10	egyptian(3/7)	VDR(3,7)	sum = 3/7	3/7	PASS
11	egyptian(5/11)	VDR(5,11)	sum = 5/11	5/11	PASS
12	egyptian(7/15)	VDR(7,15)	sum = 7/15	7/15	PASS
13	egyptian(4/17)	VDR(4,17)	sum = 4/17	4/17	PASS
14	Stern-Brocot reduced	depth 3	all reduced	all reduced	PASS
15	Stern-Brocot in (0,1)	depth 3	all positive	all positive	PASS
16	Stern-Brocot ordered	depth 3	strictly ordered	strictly ordered	PASS
17	F ₅ element count	n=5	11	11	PASS
18	Farey mediant property	F ₅	lad-bcl=1 for all pairs	verified	PASS
19	$\varphi(1)$	n=1	1	1	PASS
20	$\varphi(6)$	n=6	2	2	PASS
21	$\varphi(10)$	n=10	4	4	PASS
22	$\varphi(12)$	n=12	4	4	PASS
23	$\varphi(36)$	n=36	12	12	PASS
24	$\varphi(100)$	n=100	40	40	PASS
25	H ₁₀	sum 1/k, k=1..10	7381/2520	7381/2520	PASS
26	H ₂₀ exact	sum 1/k, k=1..20	exact rational	exact rational	PASS
27	H ₁₀₀ - H ₅₀ exact	difference	exact rational	exact rational	PASS
28	2 ⁴ mod 5	Fermat	1	1	PASS
29	3 ⁴ mod 5	Fermat	1	1	PASS
30	2 ⁶ mod 7	Fermat	1	1	PASS
31	3 ⁶ mod 7	Fermat	1	1	PASS
32	2 ¹⁰ mod 11	Fermat	1	1	PASS
33	3 ¹⁰ mod 11	Fermat	1	1	PASS
34	2 ¹² mod 13	Fermat	1	1	PASS
35	3 ¹² mod 13	Fermat	1	1	PASS

#	Exercise	Input	Expected	Got	Status
36	$\sqrt{2}$ convergents monotone	CF [1;2,2,...]	improving	improving	PASS
37	e convergent[14]	CF of e, 15 terms	error $< 10^{-10}$	verified	PASS

A.2 Gym 02: Polynomial Algebra (22/23)

#	Exercise	Input	Expected	Got	Status
1	$p(0) = -5$	$2x^3 - 3x^2 + x - 5$ at 0	-5	-5	PASS
2	$p(1) = -5$	at 1	-5	-5	PASS
3	$p(2) = 1$	at 2	1	1	PASS
4	$p(-1) = -11$	at -1	-11	-11	PASS
5	$p(1/2) = -19/4$	at 1/2	-19/4	-5	FAIL
6	$(x+1)(x-1) = x^2 - 1$	product	[-1,0,1]	[-1,0,1]	PASS
7	$(x+1)^3$	product	[1,3,3,1]	[1,3,3,1]	PASS
8	distributivity	$p1 * (p2 + p3)$	$= p1p2 + p1p3$	equal	PASS
9	$(x^3 + 2x^2 - x - 2) / (x - 1)$	poly div	remainder 0	remainder 0	PASS
10	quot*div + rem = dividend	reconstruction	equal	equal	PASS
11	$(x^2 + 1) / (x - 1)$ rem	poly div	remainder 2	remainder 2	PASS
12	rational root $x=1$	$2x^3 - 3x^2 - 3x + 2$	$p(1)=0$	0	PASS
13	rational root $x=-1/2$	same poly	$p(-1/2)=0$	0	PASS
14	rational root $x=2$	same poly	$p(2)=0$	0	PASS
15	Lagrange $p(0)=1$	interp	1	1	PASS
16	Lagrange $p(1)=3$	interp	3	3	PASS
17	Lagrange $p(2)=7$	interp	7	7	PASS
18	Lagrange coeffs [1,1,1]	interp	[1,1,1]	[1,1,1]	PASS
19	rational Lagrange $p(1/2)=1/3$	interp	1/3	1/3	PASS
20	rational Lagrange $p(1/3)=1/5$	interp	1/5	1/5	PASS

#	Exercise	Input	Expected	Got	Status
21	rational Lagrange $p(1/5)=1/7$	interp	1/7	1/7	PASS
22	poly gcd = x+1	$\gcd(x^2-1, x^2+2x+1)$	[1,1]	[1,1]	PASS
23	Cayley-Hamilton	$M^2-5M+5I = 0$	zero matrix	zero matrix	PASS

Failure analysis (exercise 5): $p(x) = 2x^3 - 3x^2 + x - 5$ evaluated at $x=1/2$: $2(1/8) - 3(1/4) + 1/2 - 5 = 1/4 - 3/4 + 1/2 - 5 = -5$. The test expected -19/4. VDR computed -5 correctly. The expected value was wrong.

A.3 Gym 03: Continued Fractions (26/31)

#	Exercise	Input	Expected	Got	Status
1	$1/2 \rightarrow CF$	VDR(1,2)	[0;2]	[0;2]	PASS
2	$3/7 \rightarrow CF$	VDR(3,7)	[0;2,3]	[0;2,3]	PASS
3	$22/7 \rightarrow CF$	VDR(22,7)	[3;7]	[3;7]	PASS
4	$355/113 \rightarrow CF$	VDR(355,113)	[3;7,16]	[3;7,16]	PASS
5	$1/1 \rightarrow CF$	VDR(1,1)	[1]	[1]	PASS
6	roundtrip 1/2	CF→VDR	1/2	1/2	PASS
7	roundtrip 3/7	CF→VDR	3/7	3/7	PASS
8	roundtrip 22/7	CF→VDR	22/7	22/7	PASS
9	roundtrip 355/113	CF→VDR	355/113	355/113	PASS
10	roundtrip 1/1	CF→VDR	1	1	PASS
11	final convergent = 355/113	convergents	355/113	355/113	PASS
12	convergent 0 best approx	bound check	$\leq$ bound	$\leq$ bound	PASS
13	convergent 1 best approx	bound check	$\leq$ bound	$\leq$ bound	PASS
14	rebase 355/113 to 7	rebase	preserves value	355/113	PASS
15	rebase 355/113 to 22	rebase	preserves value	355/113	PASS
16	rebase 355/113 to 100	rebase	preserves value	355/113	PASS
17	rebase 355/113 to 113	rebase	preserves value	355/113	PASS
18	rebase 355/113 to 226	rebase	preserves value	355/113	PASS

#	Exercise	Input	Expected	Got	Status
19	most common CF coeff is 1	Gauss- Kuzmin	count[1] > count[2]	verified	PASS
20	SB path 1/3 VDR exact	Stern- Brocot	1/3	1/3	PASS
21	SB path 2/5 VDR exact	Stern- Brocot	2/5	2/5	PASS
22	SB path 3/7 VDR exact	Stern- Brocot	3/7	3/7	PASS
23	SB path 5/8 VDR exact	Stern- Brocot	5/8	5/8	PASS
24	SB path 7/11 VDR exact	Stern- Brocot	7/11	7/11	PASS
25	$\sqrt{2}$ period	periodic CF	period 1	period 1	PASS
26	$\sqrt{3}$ period	periodic CF	period 2	period 2	PASS
27	$\sqrt{2}$ convergent error	last convergent	< 0.01	> 0.01	FAIL
28	$\sqrt{3}$ convergent error	last convergent	< 0.01	> 0.01	FAIL
29	$\sqrt{5}$ convergent error	last convergent	< 0.01	> 0.01	FAIL
30	$\sqrt{7}$ convergent error	last convergent	< 0.01	> 0.01	FAIL
31	$\sqrt{10}$ convergent error	last convergent	< 0.01	> 0.01	FAIL

#	Exercise	Input	Expected	Got	Status
	Failure 9/4 - 2	$= 1/4 =$			
	anal-	$0.25 > 0.01.$			
	ysis	The fix:			
	(ex-	extend the			
	er-	periodic			
	cises	CF to more			
	27-	repetitions			
	31):	before			
	The	computing			
	period-	the final			
	finding	convergent.			
	func-	For			
	tion	example,			
	re-	$[1; 2, 2, 2,$			
	turns	$2, 2]$ (5			
	only	repetitions)			
	the	gives			
	non-	convergent			
	repeating	$99/70$, and			
	ini-	$(99/70)^2 - 2$			
	tial	$= 1/4900 <$			
	seg-	$0.01.$ The			
	ment	period			
	plus	detection is			
	one	correct.			
	pe-	The			
	riod	convergent			
	of	quality			
	the	check			
	CF.	needs more			
	For	terms.			
	$\sqrt{2} =$				
	$[1; 2,$				
	$2, 2,$				
	$\dots]$				
	with				
	pe-				
	riod				
	1, it				
	re-				
	turns				
	$[1;$				
	$2]$ —				
	only				
	2 co-				
	effi-				
	cients.				
	The				
	con-				
	ver-				
	gent				
	from				
	$[1;$				
	$2]$ is				

A.4 Gym 04: Matrix Decomposition (13/13)

#	Exercise	Input	Expected	Got	Status
1	LU: $L*U = A$	3×3 integer	A	A	PASS
2	L lower triangular	L matrix	diag=1, upper=0	verified	PASS
3	U upper triangular	U matrix	lower=0	verified	PASS
4	LU rational reconstruct	3×3 rational	exact	exact	PASS
5	LU solve $Ax=b$	[1,2,3]	exact x	exact x	PASS
6	LU agrees with Cramer	same system	same x	same x	PASS
7	$PA = LU$	pivoted matrix	exact	exact	PASS
8	$F(10) = 55$	matrix power	55	55	PASS
9	$F(20) = 6765$	matrix power	6765	6765	PASS
10	$F(30) = 832040$	matrix power	832040	832040	PASS
11	$F(15)^2 + F(16)^2 = F(31)$	Fibonacci identity	equal	equal	PASS
12	Gram-Schmidt orthogonal	3 vectors	all dots = 0	all dots = 0	PASS
13	matrix exp deterministic	10 terms, twice	identical	identical	PASS

A.5 Gym 05: Recursive Sequences (15/15)

#	Exercise	Input	Expected	Got	Status
1	Fibonacci 0–19	F(0)..F(19)	known values	all match	PASS
2	Cassini's identity	n=2..17	$(-1)^n$	all match	PASS
3	$F(n+1)/F(n)$ exact	ratios	exact rationals	exact	PASS
4	Lucas 0–14	L(0)..L(14)	known values	all match	PASS
5	$L(n)^2 - 5F(n)^2 = 4(-1)^n$	n=2..12	identity	verified	PASS
6	Catalan 0–9	C(0)..C(9)	known values	all match	PASS

#	Exercise	Input	Expected	Got	Status
7	Catalan recurrence	$n=1..7$	identity	verified	PASS
8	$B(0)=1$	Bernoulli	1	1	PASS
9	$B(1)=-1/2$	Bernoulli	-1/2	-1/2	PASS
10	$B(2)=1/6$	Bernoulli	1/6	1/6	PASS
11	$B(12)=-691/2730$	Bernoulli	-691/2730	-691/2730	PASS
12	fn fib(0..15)	functional	matches direct	all match	PASS
13	Tribonacci 0–15	$T(0)..T(15)$	known values	all match	PASS
14	rational recurrence	$a(0)..a(14)$	$3 \cdot 2^{(1-n)}$	all match	PASS
15	Bernoulli $B(0)..B(12)$ all	13 values	known fractions	all match	PASS

A.6 Gym 06: Combinatorics (31/31)

#	Exercise	Input	Expected	Got	Status
1	$C(7,3)$	binom	35	35	PASS
2	$C(10,5)$	binom	252	252	PASS
3	$C(20,10)$	binom	184756	184756	PASS
4	Pascal's rule $n=2..14$	all valid k	identity	verified	PASS
5	Vandermonde $C(12,6)$	sum formula	equal	equal	PASS
6	$S(3,2)=3$	Stirling	3	3	PASS
7	$S(4,2)=7$	Stirling	7	7	PASS
8	$S(4,3)=6$	Stirling	6	6	PASS
9	$S(5,2)=15$	Stirling	15	15	PASS
10	$S(5,3)=25$	Stirling	25	25	PASS
11	$S(5,4)=10$	Stirling	10	10	PASS
12	$S(6,3)=90$	Stirling	90	90	PASS
13	$B(0)=1$	Bell	1	1	PASS
14	$B(1)=1$	Bell	1	1	PASS
15	$B(2)=2$	Bell	2	2	PASS
16	$B(3)=5$	Bell	5	5	PASS
17	$B(4)=15$	Bell	15	15	PASS
18	$B(5)=52$	Bell	52	52	PASS
19	$B(6)=203$	Bell	203	203	PASS
20	$B(7)=877$	Bell	877	877	PASS
21	$D(0)=1$	derangement	1	1	PASS
22	$D(1)=0$	derangement	0	0	PASS

#	Exercise	Input	Expected	Got	Status
23	D(2)=1	derangement	1	1	PASS
24	D(3)=2	derangement	2	2	PASS
25	D(4)=9	derangement	9	9	PASS
26	D(5)=44	derangement	44	44	PASS
27	D(6)=265	derangement	265	265	PASS
28	D(7)=1854	derangement	1854	1854	PASS
29	multinomial(10;3,3,4)=4200	multinomial	4200	4200	PASS
30	trinomial sum = $3^4=81$	multinomial theorem	81	81	PASS
31	Catalan GF exact rational	gen function	denom > 1	verified	PASS

A.7 Gym 07: Signal Processing (11/11)

#	Exercise	Input	Expected	Got	Status
1	conv [1,2,3]*[1,1]	linear conv	[1,3,5,3]	[1,3,5,3]	PASS
2	rational conv first elem	$[1/2, 1/3, 1/5] * [1/7, 1/11]$	$1/74$	1/14	PASS
3	autocorr symmetric	[1,2,1]	symmetric	symmetric	PASS
4	autocorr peak at center	[1,2,1]	peak	peak	PASS
5	MA(3) arithmetic seq	[0..9]	[1,2,3,...,8]	verified	PASS
6	rational MA exact	harmonic signal	exact fractions	exact	PASS
7	$H(2) = 7/12$	z-transform	7/12	7/12	PASS
8	$H(1) = 1$	DC gain	1	1	PASS
9	Toeplitz*signal = conv	matrix conv	equal	equal	PASS
10	IIR a=1/2: $y[n]=(1/2)^n$	impulse response	$(1/2)^n$	all match	PASS
11	IIR a=2/3: $y[n]=(2/3)^n$	impulse response	$(2/3)^n$	all match	PASS

A.8 Gym 08: Computational Geometry (19/19)

#	Exercise	Input	Expected	Got	Status
1	$y=x \cap y=-x+2$	line intersect	(1,1)	(1,1)	PASS
2	rational intersection exact	rational lines	exact rational	exact	PASS
3	unit square area	Shoelace	1	1	PASS
4	3-4-5 triangle area	Shoelace	6	6	PASS
5	rational triangle area	Shoelace	exact rational	exact	PASS
6	centroid bary (1/3,1/3,1/3)	barycentric	(1/3,1/3,1/3)	exact	PASS
7	vertex a bary (1,0,0)	barycentric	(1,0,0)	exact	PASS
8	midpoint bc bary (0,1/2,1/2)	barycentric	(0,1/2,1/2)	exact	PASS
9	(2,2) inside triangle	point-in-tri	True	True	PASS
10	(6,6) outside triangle	point-in-tri	False	False	PASS
11	(5,0) on edge	point-in-tri	True	True	PASS
12	(1/3,0) on edge exact	point-in-tri	True	True	PASS
13	$AB^2=9$	dist squared	9	9	PASS
14	$AC^2=16$	dist squared	16	16	PASS
15	$BC^2=25$	dist squared	25	25	PASS
16	Pythagorean $AB^2+AC^2=BC^2$	identity	25	25	PASS
17	rational dist ² exact	rational points	exact	exact	PASS
18	circumcenter = (3,4)	right triangle	(3,4)	(3,4)	PASS
19	circumcenter equidistant	3 distances	all equal	all equal	PASS

A.9 Gym 09: Differential Equations (10/10)

#	Exercise	Input	Expected	Got	Status
1	Euler $y(1)=(11/10)^{10}$	$dy/dx=y$	$(11/10)^{10}$	exact match	PASS
2	decay $y(2)=(9/10)^{20}$	$dy/dx=-y$	$(9/10)^{20}$	exact match	PASS
3	RK4 > Euler accuracy	$dy/dx=y$	RK4 err < Euler err	verified	PASS
4	RK4 exact rational	result type	rational	rational	PASS
5	matrix exp exact rational	e^A partial	all rational	all rational	PASS
6	Picard 3 = $1/k!$	Taylor coeffs	$1/k!$	all match	PASS
7	Picard 5 = $1/k!$	Taylor coeffs	$1/k!$	all match	PASS
8	Picard 8 = $1/k!$	Taylor coeffs	$1/k!$	all match	PASS
9	Picard e approx < 10^{-5}	8 terms	error < 10^{-5}	verified	PASS
10	Lotka-Volterra all exact	200 steps	all rational	all rational	PASS

A.10 Gym 10: Optimization (8/8)

#	Exercise	Input	Expected	Got	Status
1	Newton quadratic 1 step	x^2-4x+7	$x=2$ in 1 step	2	PASS
2	Newton cubic converges	x^3-6x^2+9x+1	diffs decrease	verified	PASS
3	GD approaches (0,0)	x^2+2y^2	near origin	near origin	PASS
4	GD all exact rationals	20 steps	all rational	all rational	PASS
5	simplex feasible	LP	all constraints	satisfied	PASS
6	simplex exact rationals	solution	all rational	all rational	PASS
7	bisection all exact	30 midpoints	all rational	all rational	PASS
8	bisection error < 10^{-8}	30 steps	< 10^{-8}	verified	PASS

A.11 Gym 11: Probability (13/13)

#	Exercise	Input	Expected	Got	Status
1	P(disease +) exact	Bayes	exact rational	exact	PASS
2	probabilities sum to 1	Bayes	1	1	PASS
3	Markov steady = [2/3,1/3]	weather	[2/3,1/3]	[2/3,1/3]	PASS
4	ruin P(k=1) = 4/5	gambler	4/5	4/5	PASS
5	ruin P(k=2) = 3/5	gambler	3/5	3/5	PASS
6	ruin P(k=3) = 2/5	gambler	2/5	2/5	PASS
7	ruin P(k=4) = 1/5	gambler	1/5	1/5	PASS
8	E[die] = 7/2	expected value	7/2	7/2	PASS
9	Var[die] = 35/12	variance	35/12	35/12	PASS
10	Binom PMF sums to 1	B(10,1/3)	1	1	PASS
11	E[Binom] = 10/3	expected value	10/3	10/3	PASS
12	Bayesian posterior exact	3 updates	exact rational	exact	PASS
13	final posterior = 6/7	odds form	6/7	6/7	PASS

A.12 Gym 12: Cryptographic Primitives (37/37)

#	Exercise	Input	Expected	Got	Status
1	$2^6 \bmod 7 = 1$	Fermat	1	1	PASS
2	$3^6 \bmod 7 = 1$	Fermat	1	1	PASS
3	$5^6 \bmod 7 = 1$	Fermat	1	1	PASS
4	$2^{10} \bmod 11 = 1$	Fermat	1	1	PASS
5	$3^{10} \bmod 11 = 1$	Fermat	1	1	PASS
6	$5^{10} \bmod 11 = 1$	Fermat	1	1	PASS
7	$2^{12} \bmod 13 = 1$	Fermat	1	1	PASS
8	$3^{12} \bmod 13 = 1$	Fermat	1	1	PASS
9	$5^{12} \bmod 13 = 1$	Fermat	1	1	PASS
10	$2^{16} \bmod 17 = 1$	Fermat	1	1	PASS
11	$3^{16} \bmod 17 = 1$	Fermat	1	1	PASS
12	$5^{16} \bmod 17 = 1$	Fermat	1	1	PASS
13	$2^{22} \bmod 23 = 1$	Fermat	1	1	PASS
14	$3^{22} \bmod 23 = 1$	Fermat	1	1	PASS

#	Exercise	Input	Expected	Got	Status
15	$5^{22} \bmod 23 = 1$	Fermat	1	1	PASS
16	$2^{28} \bmod 29 = 1$	Fermat	1	1	PASS
17	$3^{28} \bmod 29 = 1$	Fermat	1	1	PASS
18	$5^{28} \bmod 29 = 1$	Fermat	1	1	PASS
19	$2^{30} \bmod 31 = 1$	Fermat	1	1	PASS
20	$3^{30} \bmod 31 = 1$	Fermat	1	1	PASS
21	$5^{30} \bmod 31 = 1$	Fermat	1	1	PASS
22	$\gcd(35,15) = 5$	ext GCD	5	5	PASS
23	Bezout identity	$35x+15y=5$	holds	holds	PASS
24	$3^{-1} \bmod 7: 3 \times$ $5 \bmod 7=1$	mod inv	1	1	PASS
25	$5^{-1} \bmod 11: 5 \times$ $9 \bmod 11=1$	mod inv	1	1	PASS
26	$7^{-1} \bmod 13: 7 \times$ $2 \bmod 13=1$	mod inv	1	1	PASS
27	CRT x mod 3 = 2	CRT verify	2	2	PASS
28	CRT x mod 5 = 3	CRT verify	3	3	PASS
29	CRT x mod 7 = 2	CRT verify	2	2	PASS
30	RSA $e*d \bmod \varphi$ $= 1$	key gen	1	1	PASS
31	RSA roundtrip msg=42	encrypt/decrypt	42	42	PASS
32	RSA roundtrip msg=7	encrypt/decrypt	7	7	PASS
33	RSA roundtrip msg=100	encrypt/decrypt	100	100	PASS
34	RSA roundtrip msg=1234	encrypt/decrypt	1234	1234	PASS
35	RSA roundtrip msg=3000	encrypt/decrypt	3000	3000	PASS
36	$\text{dlog } 2^x \equiv 8 \bmod 13: x=3$	baby-giant	3	3	PASS
37	$\text{dlog } 3^x \equiv 9 \bmod 17$ verify	baby-giant	$3^x \bmod 17=9$	verified	PASS

A.13 Gym 13: Symbolic Algebra (20/20)

#	Exercise	Input	Expected	Got	Status
1	PFD A = -1	$1/((x-1)(x-2))$	-1	-1	PASS
2	PFD B = 1	$1/((x-1)(x-2))$	1	1	PASS

#	Exercise	Input	Expected	Got	Status
3	PFD verified at x=3	evaluation	$1/2 = 1/2$	equal	PASS
4	PFD complex verified x=5	3 terms	equal	equal	PASS
5	rational function sum x=3	$1/(x-1)+1/(x+1)$	$3/4$	$3/4$	PASS
6	$S_1(10) = 55$	power sum	55	55	PASS
7	$S_2(10) = 385$	power sum	385	385	PASS
8	$S_3(10) = S_1(10)^2$	power sum	3025	3025	PASS
9	$S_1(50)$	power sum formula	equal	equal	PASS
10	$S_2(50)$	power sum formula	equal	equal	PASS
11	$S_3(50) = S_1(50)^2$	power sum	equal	equal	PASS
12	$S_1(100)$	power sum formula	equal	equal	PASS
13	$S_2(100)$	power sum formula	equal	equal	PASS
14	$S_3(100) = S_1(100)^2$	power sum	equal	equal	PASS
15	$d/dx(x^3+2x^2-x+5)$	poly deriv	$3x^2+4x-1$	exact	PASS
16	$\int(\text{derivative}) = \text{original}$	integration	equal	equal	PASS
17	$\int_0^1 x^2 dx = 1/3$	definite	$1/3$	$1/3$	PASS
18	$\int_0^2 x^3 dx = 4$	definite	4	4	PASS
19	$\int_1^3 (3x^2+2x+1) dx = 36$	definite	36	36	PASS
20	$\int \{1/3\}^{\{2/3\}} x^2 dx$	rational bounds	exact	exact	PASS

Appendix B. Bernoulli Numbers Computed by VDR

n	B(n) exact	Decimal approx
0	1	1.0
1	-1/2	-0.5
2	1/6	0.16667
3	0	0.0
4	-1/30	-0.03333
5	0	0.0
6	1/42	0.02381
7	0	0.0
8	-1/30	-0.03333
9	0	0.0
10	5/66	0.07576
11	0	0.0
12	-691/2730	-0.25311

All computed by the recursive formula using exact VDR arithmetic. $B(12) = -691/2730$ is the first Bernoulli number with a numerator not equal to ± 1 or 0. This fraction appears in the theory of modular forms and in the Riemann zeta function at even positive integers.

VDR computes it from the recursion without any special-case handling.

Appendix C. Fibonacci Identity Verification Table

C.1 Cassini's Identity: $F(n-1) \cdot F(n+1) - F(n)^2 = (-1)^n$

n	F(n-1)	F(n)	F(n+1)	F(n-1)·F(n+1)	F(n) ²	Difference	(-1) ⁿ	Match
2	1	1	2	2	1	1	1	✓
3	1	2	3	3	4	-1	-1	✓
4	2	3	5	10	9	1	1	✓
5	3	5	8	24	25	-1	-1	✓
6	5	8	13	65	64	1	1	✓
7	8	13	21	168	169	-1	-1	✓
8	13	21	34	442	441	1	1	✓
9	21	34	55	1155	1156	-1	-1	✓
10	34	55	89	3026	3025	1	1	✓

Every identity verified by exact VDR integer multiplication.

C.2 Lucas-Fibonacci Identity: $L(n)^2 - 5 \cdot F(n)^2 = 4 \cdot (-1)^n$

n	L(n)	F(n)	L(n) ²	5·F(n) ²	Difference	4·(-1) ⁿ	Match
2	3	1	9	5	4	4	✓
3	4	2	16	20	-4	-4	✓
4	7	3	49	45	4	4	✓
5	11	5	121	125	-4	-4	✓
6	18	8	324	320	4	4	✓
7	29	13	841	845	-4	-4	✓
8	47	21	2209	2205	4	4	✓

Appendix D. Harmonic Number Exact Values

n	H n exact fraction	Decimal approx	Denominator digit count
1	1	1.0	1
2	3/2	1.5	1
5	137/60	2.2833	2
10	7381/2520	2.9290	4

n	H n exact fraction	Decimal approx	Denominator digit count
15	1195757/360360	3.3182	6
20	55835135/15519504	3.5977	8
50	(exact, ~40 digit denom)	4.4992	~40
100	(exact, ~85 digit denom)	5.1874	~85

Every H_n is computed as a single exact VDR rational by summing $1/k$ for $k=1..n$. The denominators grow as $\text{lcm}(1,2,\dots,n)$. Float computation of H_{100} loses precision in the last digits due to adding small terms ($1/100$) to large partial sums (≈ 5). VDR has no such issue — every addition is exact.

Appendix E. Egyptian Fraction Decompositions

Fraction	Decomposition	Term count	Verified sum
2/3	1/2 + 1/6	2	2/3 ✓
3/7	1/3 + 1/11 + 1/231	3	3/7 ✓
5/11	1/3 + 1/9 + 1/99	3	5/11 ✓
7/15	1/3 + 1/8 + 1/120	3	7/15 ✓
4/17	1/5 + 1/29 + 1/1233 + 1/3039345	4	4/17 ✓

The greedy algorithm produces unit fractions with rapidly growing denominators. $4/17$ requires a denominator of 3,039,345 in its fourth term. VDR computes and verifies the sum exactly.

Appendix F. Farey Sequence F_5

Index	Fraction	VDR form
0	0/1	[0, 1, 0]
1	1/5	[1, 5, 0]
2	1/4	[1, 4, 0]
3	1/3	[1, 3, 0]
4	2/5	[2, 5, 0]
5	1/2	[1, 2, 0]
6	3/5	[3, 5, 0]
7	2/3	[2, 3, 0]
8	3/4	[3, 4, 0]
9	4/5	[4, 5, 0]
10	1/1	[1, 1, 0]

Index	Fraction	VDR form
11 elements. Every consecutive pair (a/b, c/d) satisfies	ad - bc	= 1 exactly.

Appendix G. IIR Filter Impulse Response

G.1 Filter coefficient a = 1/2

n	y[n] exact	Decimal
0	1	1.0
1	1/2	0.5
2	1/4	0.25
3	1/8	0.125
4	1/16	0.0625
5	1/32	0.03125
6	1/64	0.015625
7	1/128	0.007813
8	1/256	0.003906
9	1/512	0.001953

G.2 Filter coefficient a = 2/3

n	y[n] exact	Decimal
0	1	1.0
1	2/3	0.66667
2	4/9	0.44444
3	8/27	0.29630
4	16/81	0.19753
5	32/243	0.13169
6	64/729	0.08779
7	128/2187	0.05853

Every output sample is an exact VDR rational. In float-based DSP, these values would accumulate rounding error over long filter runs. In VDR, $(2/3)^{100}$ is computed as an exact fraction $2^{100/3} 100$ with no loss.

Appendix H. Gambler's Ruin Exact Solutions

H.1 Symmetric walk $p = q = 1/2$, $N = 5$

Start k	P(ruin at 0)	Expected 1-k/N
1	4/5	4/5 ✓
2	3/5	3/5 ✓
3	2/5	2/5 ✓
4	1/5	1/5 ✓

Computed by solving a tridiagonal linear system using exact VDR Cramer's rule.

Appendix I. Binomial PMF: $B(10, 1/3)$

k	P(X=k) exact	Decimal
0	1024/59049	0.01734
1	5120/59049	0.08671
2	11520/59049	0.19509
3	15360/59049	0.26012
4	13440/59049	0.22761
5	8064/59049	0.13656
6	3360/59049	0.05690
7	960/59049	0.01626
8	180/59049	0.00305
9	20/59049	0.00034
10	1/59049	0.00002
Sum	59049/59049	1.00000

The PMF sums to exactly 1. The common denominator $59049 = 3^{10}$ arises from $(1 - 1/3)^{(10-k)}$ terms. Every probability is an exact VDR rational.

Appendix J. RSA Toy Example Complete Trace

Parameter	Value
p	61
q	53
$n = p \cdot q$	3233
$\varphi(n) = (p-1)(q-1)$	3120

Parameter	Value
e (public)	17
d (private) = $e^{-1} \bmod \varphi(n)$	2753
$e \cdot d \bmod \varphi(n)$	1 ✓
Message	Encrypted ($m^e \bmod n$)
7	computed
42	computed
100	computed
1234	computed
3000	computed

Every step uses exact VDR modular arithmetic. The private key $d = 2753$ is computed via the extended Euclidean algorithm, verified by $e \cdot d = 17 \cdot 2753 = 46801 \equiv 1 \pmod{3120}$.

Appendix K. Chaotic Map Denominator Growth

K.1 Logistic Map $r=4$, $x_0=1/3$

Step	Approximate denominator digit count
1	1
2	2
3	3
4	5
5	9
6	17
7	33
8	65
9	129
10	258
15	~8,000
20	~260,000
25	~8,000,000
30	~260,000,000

Denominator digit count approximately doubles at each step, consistent with the Lyapunov exponent $\ln(2) \approx 0.693$. Each iterate is exact. The computational cost grows as $O(n^2)$ in denominator digit count for each multiplication, making step 20+ impractical on consumer hardware.

K.2 Tent Map on $1/7$

Step	Value	Denominator
1	2/7	7
2	4/7	7
3	6/7	7
4	2/7	7
5	4/7	7
6	6/7	7

Period 3. Denominator stays at 7 forever. The tent map on p/q with $\gcd(p,q)=1$ has periodic orbits with denominators that divide q . No growth. No computational difficulty. This is the exact opposite of the chaotic logistic map — periodic rational orbits are free.

K.3 Arnold Cat Map on (1/7, 3/11)

Period 40 computed exactly. The denominator stays bounded by $\text{lcm}(7,11) = 77$ throughout the orbit because the cat map on the rational torus $\mathbb{Z}/q\mathbb{Z} \times \mathbb{Z}/q\mathbb{Z}$ is periodic. Every coordinate has denominator dividing 77. All 40 points are exact rational pairs. Float arithmetic could not reliably detect this period because modular reduction of floats accumulates error, eventually placing the orbit on the wrong side of a boundary.

Appendix L. Euler and RK4 Comparison for $dy/dx = y$

L.1 Euler Method, $h = 1/10$

Step	x	y exact VDR	y decimal	e^x
0	0	1	1.0	1.0
2	1/5	121/100	1.21	1.2214
4	2/5	14641/10000	1.4641	1.4918
6	3/5	1771561/1000000	1.77156	1.8221
8	4/5	214358881/100000000	2.14359	2.2255
10	1	25937424601/10000000000	2.59374	2.71828
Euler error at x=1:	-	2.59374 $\approx$ 0.125. Every VDR value is $(11/10)^n$ exactly.		

L.2 RK4 Method, $h = 1/4$

Step	x	y decimal	e^x	Error
0	0	1.0	1.0	0
1	1/4	1.28394...	1.28403	$\sim 9 \times 10^{-5}$
2	1/2	1.64851...	1.64872	$\sim 2 \times 10^{-4}$

Step	x	y decimal	e^x	Error
3	3/4	2.11681...	2.11700	$\sim 2 \times 10^{-4}$
4	1	2.71735...	2.71828	$\sim 9 \times 10^{-4}$

RK4 error at $x=1$: $\approx 9 \times 10^{-4}$, about $140 \times$ better than Euler despite using only 4 steps vs 10. Both results are exact VDR rationals — the error is method error (discretization), not arithmetic error.

Appendix M. Picard Iteration Taylor Coefficients

M.1 Picard iteration for $dy/dx = y$, $y(0) = 1$

Iteration	Polynomial coefficients	Matches $1/k!$ through degree
1	[1, 1]	degree 1
2	[1, 1, 1/2]	degree 2
3	[1, 1, 1/2, 1/6]	degree 3
4	[1, 1, 1/2, 1/6, 1/24]	degree 4
5	[1, 1, 1/2, 1/6, 1/24, 1/120]	degree 5
8	[1, 1, 1/2, 1/6, 1/24, 1/120, 1/720, 1/5040, 1/40320]	degree 8

Every coefficient is the exact reciprocal of a factorial. Sum at $x=1$ after 8 iterations: $109601/40320 = 2.71828\dots$ approximating e to 5 decimal places. The fraction $109601/40320$ is exact — it is the 8th partial sum of the Taylor series for e , represented as a single VDR rational.

Appendix N. Power Sum Verification

n	$S_1(n) = \sum k$	$n(n+1)/2$	Match	$S_2(n) = \sum k^2$	$n(n+1)(2n+1)/6$	Match	$S_3(n) = \sum k^3$	$[n(n+1)/2]^2$	Match
10	55	55	✓	385	385	✓	3025	3025	✓
50	1275	1275	✓	42925	42925	✓	1625625	1625625	✓
100	5050	5050	✓	338350	338350	✓	25502500	25502500	✓

Every sum is computed by iterating VDR addition and multiplication, then compared against the closed-form formula. All 9 comparisons are exact equality — not approximate.

Appendix O. Exact Definite Integrals via Antiderivative

Integrand	Bounds	Antiderivative	Result	Expected	Match
x^2	[0, 1]	$x^3/3$	1/3	1/3	✓
x^3	[0, 2]	$x^4/4$	4	4	✓
$3x^2+2x+1$	[1, 3]	x^3+x^2+x	36	36	✓
x^2	[1/3, 2/3]	$x^3/3$	7/81	7/81	✓

These are exact symbolic integrals — the antiderivative is computed as a VDR polynomial, evaluated at the exact rational endpoints, and the difference is an exact VDR rational. This is not numerical integration. It is exact algebraic evaluation of the fundamental theorem of calculus for polynomial integrands with rational coefficients and rational bounds.

Appendix P. Failure Analysis Summary

Gym	Exercise	Expected	Got	Root Cause	Fix
02	p(1/2)	-19/4	-5	Wrong expected value	Change expected to -5
03	$\sqrt{2}$ convergent	error < 0.01	error = 0.25	Too few CF terms	Extend periodic CF before convergent
03	$\sqrt{3}$ convergent	error < 0.01	error > 0.01	Too few CF terms	Same fix
03	$\sqrt{5}$ convergent	error < 0.01	error > 0.01	Too few CF terms	Same fix
03	$\sqrt{7}$ convergent	error < 0.01	error > 0.01	Too few CF terms	Same fix
03	$\sqrt{10}$ convergent	error < 0.01	error > 0.01	Too few CF terms	Same fix
15	tent map period 6	period 6	period 3	Wrong expected period	Change expected to 3

All 7 failures (6 in completed gyms + 1 in partial gym 15) are test-authoring errors. Zero are VDR computation errors. The VDR library computed the correct exact answer in every case.

Appendix Q. Domain Suitability Matrix

Domain	Exact?	Fast?	Advantage over float	Advantage over CAS
Integer arithmetic	✓	✓	Equal (both exact)	Simpler
Rational arithmetic	✓	✓	Exact vs approximate	Simpler, faster
Number theory	✓	✓	Exact modular ops	Comparable
Combinatorics	✓	✓	Exact large integers	Comparable
Polynomial algebra	✓	✓	Exact coefficients	Simpler for rationals
Linear algebra (small)	✓	✓	Zero drift, exact inverse	Comparable
Linear algebra (large)	✓	Slow	Zero drift	CAS faster algorithms
Computational geometry	✓	✓	No robustness failures	Simpler
Probability	✓	✓	PMF sums to exactly 1	Simpler
Cryptography	✓	✓	Required (float cannot do)	Comparable
Signal processing	✓	✓	Exact filter coefficients	Simpler
Discrete ODE (stable)	✓	✓	Zero arithmetic drift	Simpler
Discrete ODE (chaotic)	✓	Very slow	Exact but impractical	CAS also slow
Optimization	✓	✓	Exact iterates	Comparable
Symbolic calculus (poly)	✓	✓	Exact integrals	CAS more general

Domain	Exact?	Fast?	Advantage over float	Advantage over CAS
Chaotic iteration	✓	Exponential cost	Honest vs hidden cost	Both expensive

References

[[HOWL-VDR-2-2026](#)] VDR Gym: Exact Arithmetic Across Fifteen Domains. Github: <https://github.com/ghowland/papers/tree/main/papers/VDR/HOWL-VDR-2-2026>

[[HOWL-VDR-1-2026](#)] VDR Arithmetic: Value, Decimal, Remainder. Github: <https://github.com/ghowland/papers/tree/main/papers/VDR/HOWL-VDR-1-2026>